

BLUESTOCK FINTECH
MUTUAL FUND ANALYTICS PLATFORM
Capstone Project Report

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1. Executive Summary

The Mutual Fund Analytics Platform was developed as part of the Bluestock Fintech Capstone Project to provide a centralized solution for analyzing mutual fund performance, investor behavior, and industry trends. The project addresses the challenge of fragmented mutual fund data by integrating multiple datasets, including NAV history, AUM statistics, SIP inflows, benchmark indices, portfolio holdings, and investor transactions into a unified analytical platform.

An end-to-end ETL pipeline was designed and implemented using Python and Pandas to extract, clean, transform, and load data into a SQLite database. Data quality checks were performed to handle missing values, remove duplicates, standardize formats, and ensure consistency across datasets. The processed data was then structured into fact and dimension tables to support efficient analysis and reporting.

Comprehensive exploratory data analysis was conducted to identify trends in fund performance, industry growth, investor demographics, SIP participation, and geographic investment distribution. Key financial metrics such as Annualized Return, CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR) were analyzed to evaluate fund performance and risk characteristics.

An interactive Power BI dashboard was developed to visualize industry trends, compare mutual fund performance, analyze investor behavior, and monitor SIP growth. The dashboard incorporates filters, drill-through functionality, and dynamic visualizations to support data-driven decision-making.

The project demonstrates the practical application of data engineering, financial analytics, and business intelligence tools in the mutual fund domain. By transforming raw financial data into actionable insights, the platform enables investors, analysts, and financial institutions to make informed investment decisions while improving accessibility to mutual fund analytics.

2. Project Objectives

The primary objectives achieved in this project were:

- Build an automated ETL pipeline for mutual fund datasets.
- Design and implement a relational database schema.
- Perform exploratory data analysis on NAV, AUM, SIP, and investor data.
- Compute performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- Develop an interactive Power BI dashboard.
- Analyze investor demographics and transaction behavior.
- Compare mutual fund performance against benchmark indices.
- Generate business insights and recommendations.

3. Data Sources

This project utilized ten datasets obtained from AMFI-related public sources and simulated investor data.

Datasets Used:

Dataset	Description
fund_master	Mutual fund master information
nav_history	Daily NAV history
aum_by_fund_house	Quarterly AMC AUM data
monthly_sip_inflows	Industry SIP statistics
category_inflows	Category-wise net inflows
industry_folio_count	Mutual fund folio growth
scheme_performance	Fund performance metrics
investor_transactions	SIP, lumpsum and redemption transactions

portfolio_holdings	Fund holdings and sector allocation
Benchmark_indices	Nifty and BSE benchmark data

4. Tools & Technologies Used

Tool	Purpose
Python	Data processing and analytics
Pandas	Data manipulation
Numpy	Numerical computations
SQLite	Database storage
Jupyter Notebook	Analysis and development
Power BI	Dashboard creation
Git & GitHub	Version control
Matplotlib/Seaborn	Data visualization

5. ETL Pipeline

Extract:

- Imported all CSV datasets using Pandas.
- Retrieved live NAV data using mfapi.in API.
- Validated AMFI scheme codes.

Transform:

- Cleaned missing values and duplicates.
- Standardized date formats.
- Forward-filled NAV values for market holidays.
- Calculated daily returns and rolling metrics.
- Normalized fund and category names.

Load:

- Loaded processed datasets into SQLite database.
- Implemented fact and dimension tables.
- Created indexed schema for efficient querying.

Database Schema:

- dim_fund
- fact_nav
- fact_transactions
- fact_performance
- fact_portfolio

6. Exploratory Data Analysis (EDA)

NAV Trends

Daily NAV data from 2022 to 2026 was analyzed to study the growth patterns of mutual fund schemes. Most equity-oriented funds exhibited a steady upward trend over the period, reflecting positive market performance and long-term wealth creation. Temporary market corrections were observed during periods of volatility; however, the majority of funds recovered and continued their growth trajectory.

AUM Analysis

Assets Under Management (AUM) data was analyzed to evaluate the growth of major fund houses. The analysis showed consistent expansion across leading AMCs, with SBI Mutual Fund maintaining the highest AUM throughout the study period. The increasing AUM trend highlights growing investor participation and confidence in mutual fund investments.

SIP Growth

Monthly SIP inflow data was examined to understand retail investor participation. The analysis revealed a continuous increase in SIP contributions over the years, culminating in record inflows during 2025. This trend indicates a strong preference for disciplined and long-term investment strategies among investors.

Investor Demographics

Investor transaction data was analyzed to identify demographic patterns. The results showed that individuals between the ages of 25 and 40 constituted the largest investor segment. Younger investors demonstrated a greater preference for SIP investments, reflecting increasing awareness of systematic wealth creation.

Geographic Distribution

State-wise transaction analysis highlighted significant regional differences in investment activity. Maharashtra, Karnataka, and Delhi recorded the highest transaction volumes and investment amounts. The findings indicate that metropolitan and economically developed regions continue to dominate mutual fund participation.

7. Fund Performance Analysis

Fund performance was evaluated using key risk and return metrics derived from NAV history and performance datasets. The analysis focused on comparing mutual funds across multiple dimensions, including returns, volatility, and benchmark performance.

Metrics Analysed

- Annualized Return
- CAGR (1 Year, 3 Year, and 5 Year)

- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Tracking Error
- Value at Risk (VaR)
- Conditional Value at Risk (CVaR)

8. Advanced Analytics

Value at Risk Analysis

Historical Value at Risk was calculated at the 95% confidence level using daily return distributions. VaR estimates the maximum expected loss under normal market conditions over a one-day horizon. Conditional VaR (CVaR), also known as Expected Shortfall, measures the average loss beyond the VaR threshold and provides a more comprehensive view of downside risk. Funds with higher VaR and CVaR values were identified as relatively more volatile and risk-prone.

Rolling Sharpe Ratio

A rolling 90-day Sharpe Ratio was computed for selected mutual fund schemes to monitor changes in risk-adjusted performance over time. This analysis helped identify periods when funds delivered superior returns relative to volatility and highlighted the consistency of fund management across different market conditions.

Investor Cohort Analysis

Investors were grouped based on the year of their first transaction, such as 2024 and 2025 cohorts. The analysis compared average SIP amounts, total investments, and preferred fund categories across cohorts. Results showed that newer investor cohorts contributed significantly to SIP growth and displayed a strong preference for equity-oriented schemes.

SIP Continuity Analysis

For investors with six or more SIP transactions, the average gap between transactions was calculated. Investors with gaps greater than 35 days were classified as "at-risk" for SIP discontinuation. This analysis can help asset management companies identify investors who may require engagement or retention efforts.

Sector Concentration Analysis

Portfolio holdings data was used to compute the Herfindahl-Hirschman Index (HHI) for sector allocations in equity funds. HHI measures the concentration of investments across sectors, where higher values indicate lower diversification. The analysis identified funds with concentrated exposure to sectors such as Financial Services or Information Technology, helping assess portfolio diversification risk.

9. Dashboard Overview

Page 1 – Industry Overview

Visualizations:

- Total Industry AUM
- SIP Inflows
- Folio Count
- Industry AUM Trend
- Top Fund Houses by AUM

Page 2 – Fund Performance

Visualizations:

- Return vs Risk Scatter Plot
- Fund Scorecard
- NAV vs Benchmark Comparison
- Fund Filters and Slicers

Page 3 – Investor Analytics

Visualizations:

- State-wise Investments
- Transaction Type Distribution
- Age Group Analysis

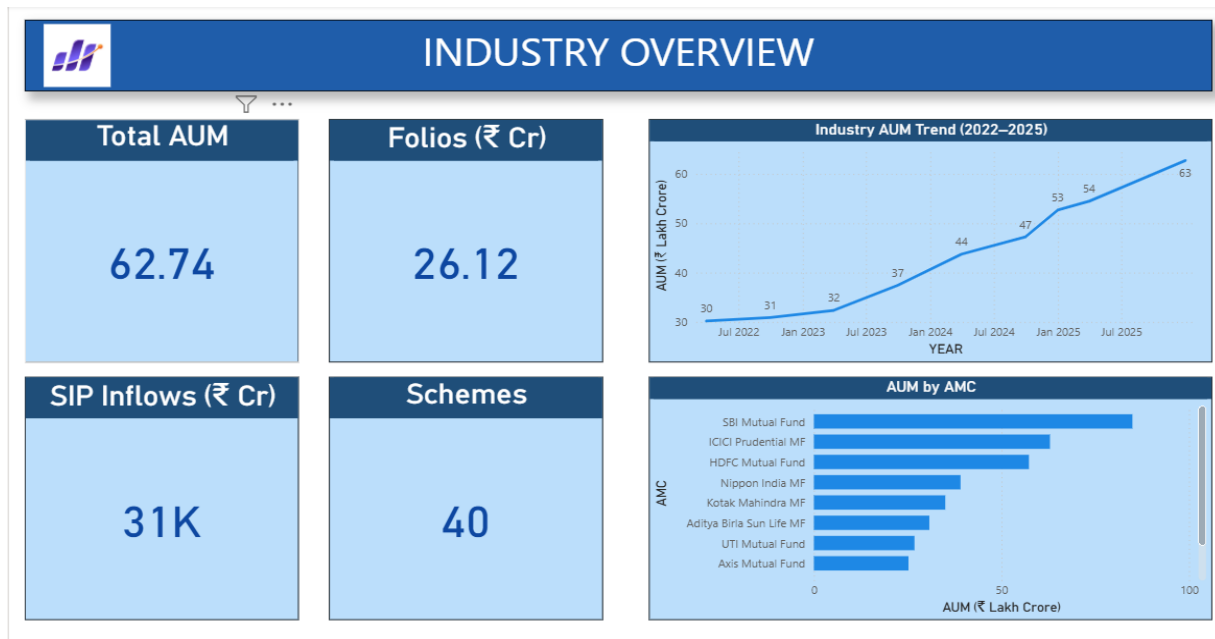
- Monthly Transaction Trends

Page 4 – SIP & Market Trends

Visualizations:

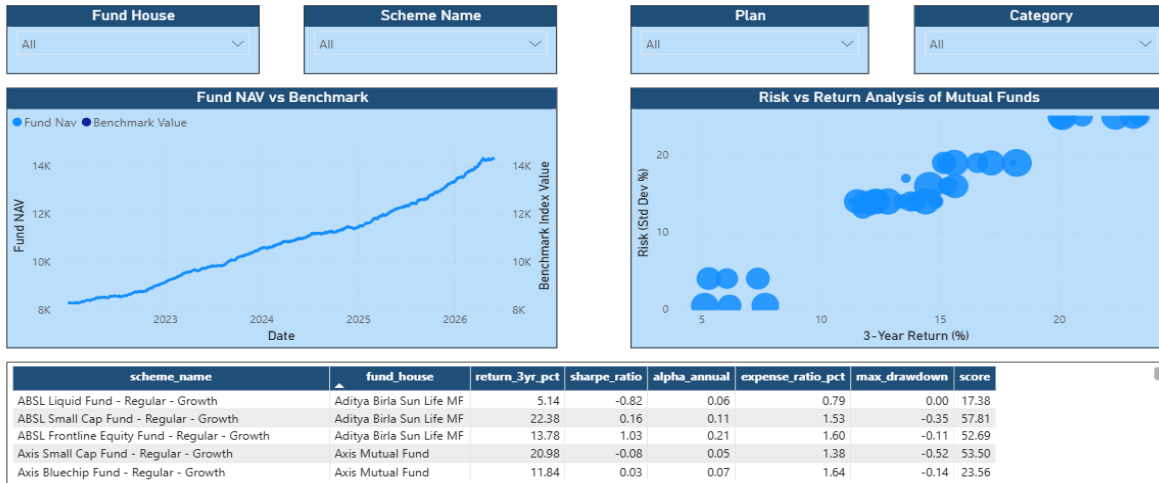
- SIP Inflows vs Nifty 50
- Category Inflow Heatmap
- Top Categories by Inflow
- SIP Account Growth

10. Dashboard Screenshots

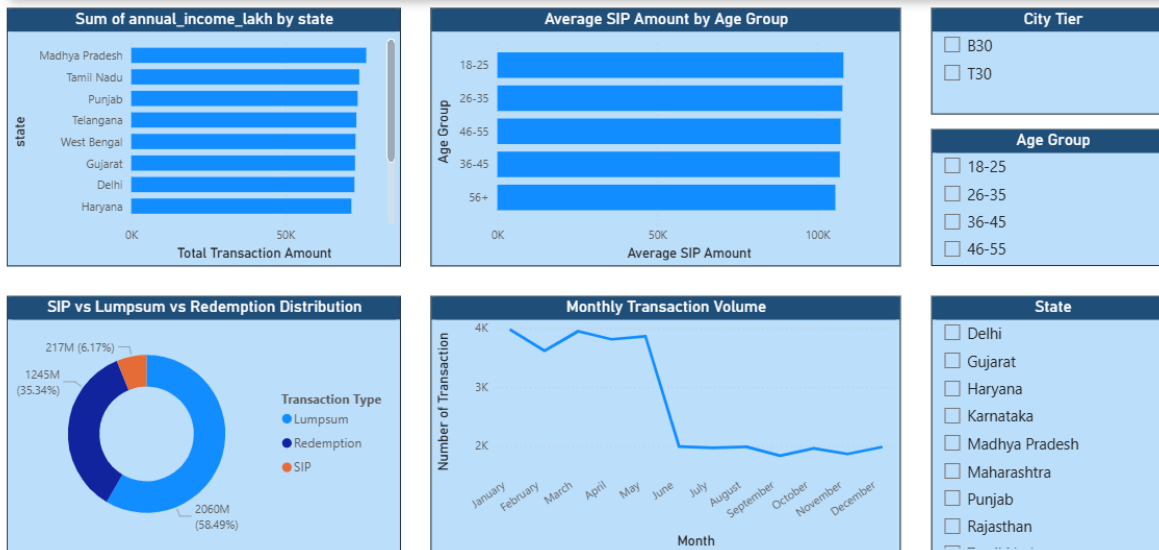




FUND PERFORMANCE

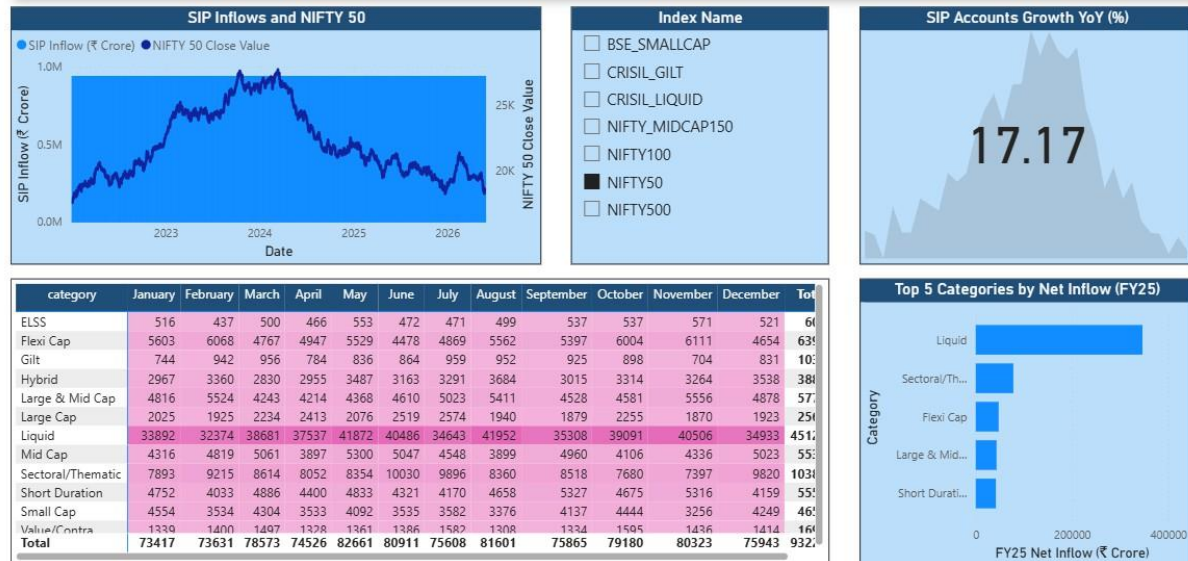


INVESTOR ANALYTICS





SIP AND MARKET TRENDS



11. Key Findings & Insights:

- Industry AUM showed strong growth between 2022 and 2025.
- SIP inflows reached record levels, indicating increasing retail participation.
- Equity funds generated higher long-term returns than debt-oriented schemes.
- Younger investors preferred SIP investments over lumpsum investments.
- Several funds demonstrated strong risk-adjusted performance based on Sharpe and Sortino ratios.

12. Recommendations

- Investors should consider risk-adjusted performance metrics such as Sharpe Ratio and Sortino Ratio alongside returns when selecting mutual funds.
- Regular SIP investments should be encouraged as they help build long-term wealth through disciplined investing.
- Fund performance should be periodically compared with benchmark indices to assess consistency and market outperformance.
- Investors should maintain a diversified portfolio across sectors and fund categories to reduce concentration risk.
- Asset management companies can leverage investor transaction analytics to identify at-risk SIP investors and improve retention strategies.

13. Limitations

- Some datasets used in the project were simulated and may not fully represent real-world investor behavior.
- Historical data availability was limited for certain schemes, which may affect long-term performance analysis.
- Risk metrics such as Sharpe Ratio, Alpha, and VaR are based on historical data and do not guarantee future performance.

14. Conclusion

The Mutual Fund Analytics Platform successfully consolidated mutual fund data from multiple sources into a centralized analytics solution. Through ETL processes, data analysis, risk metric evaluation, and Power BI dashboards, the project provided meaningful insights into fund performance, investor behavior, and industry trends.

The project demonstrated the effective use of Python, SQLite, and Power BI for financial data analysis and decision-making. Overall, the platform serves as a valuable tool for evaluating mutual funds and supports data-driven investment decisions.